1. Basic Arithmetic Formulas

Calculation	Formula
Addition	a + b
Subtraction	a - b
Multiplication	a × b
Division	a / b
Quotient	a // b
Remainder	a % b
Power	a ** b
Square	a ** 2
Cube	a ** 3

---

2. Percentage Formulas

Percentage

Percentage = (Obtained Marks / Total Marks) × 100

percentage = (obtained / total) * 100

Percentage Increase

Increase % = ((New Value − Old Value) / Old Value) × 100

Percentage Decrease

Decrease % = ((Old Value − New Value) / Old Value) × 100

Find Percentage of a Number

Value = (Percentage / 100) × Number

Example:

discount = (10 / 100) * 5000

---

3. Profit and Loss

Profit

Profit = Selling Price − Cost Price

profit = sp - cp

Loss

Loss = Cost Price − Selling Price

loss = cp - sp

Profit Percentage

Profit % = (Profit / Cost Price) × 100

Loss Percentage

Loss % = (Loss / Cost Price) × 100

---

4. Discount

Discount Amount

Discount = (Discount % / 100) × Marked Price

discount = (discount_percent / 100) * mp

Selling Price After Discount

Selling Price = Marked Price − Discount

Or directly:

Selling Price = Marked Price × (100 − Discount%) / 100

---

5. Simple Interest

Simple Interest

SI = (P × R × T) / 100

Where:

- P = Principal amount
- R = Rate of interest
- T = Time

si = (p * r * t) / 100

Total Amount

Amount = Principal + Simple Interest

amount = p + si

---

6. Average

Average

Average = Sum of Values / Number of Values

For 3 numbers:

Average = (a + b + c) / 3

average = (a + b + c) / 3

---

7. Rectangle

Area

Area = Length × Breadth

area = length * breadth

Perimeter

Perimeter = 2 × (Length + Breadth)

perimeter = 2 * (length + breadth)

---

8. Square

Area

Area = Side × Side

area = side ** 2

Perimeter

Perimeter = 4 × Side

perimeter = 4 * side

---

9. Triangle

Area

Area = ½ × Base × Height

area = 0.5 * base * height

Perimeter

Perimeter = a + b + c

perimeter = a + b + c

---

10. Circle

Use **π = 3.14** for beginner programs.

Area

Area = π × r²

area = 3.14 * radius ** 2

Circumference

Circumference = 2 × π × r

circumference = 2 * 3.14 * radius

Diameter

Diameter = 2 × Radius

diameter = 2 * radius

---

11. Cube

Volume

Volume = Side^3

volume = side ** 3

Total Surface Area

TSA = 6 × Side^2

tsa = 6 * side ** 2

---

12. Cuboid

Volume

Volume = Length × Breadth × Height

volume = length * breadth * height

Total Surface Area

TSA = 2(lb + bh + lh)

tsa = 2 * (length * breadth + breadth * height + length * height)

Total Edge Length

Total Edge = 4(l + b + h)

---

13. Speed, Distance and Time

Speed

Speed = Distance / Time

speed = distance / time

Distance

Distance = Speed × Time

distance = speed * time

Time

Time = Distance / Speed

time = distance / speed

Unit Conversion

km → m

meters = km * 1000

m → km

km = meters / 1000

---

14. Temperature

Celsius → Fahrenheit

F = (C × 9/5) + 32

fahrenheit = (celsius * 9 / 5) + 32

Fahrenheit → Celsius

C = (F − 32) × 5/9

celsius = (fahrenheit - 32) * 5 / 9

---

15. Time Conversion

Seconds → Minutes

Minutes = Seconds // 60

Remaining Seconds = Seconds % 60

minutes = seconds // 60

remaining_seconds = seconds % 60

Minutes → Hours

Hours = Minutes // 60

Remaining Minutes = Minutes % 60

hours = minutes // 60

remaining_minutes = minutes % 60

Hours → Minutes

Minutes = Hours × 60

---

16. Digit-Based Formulas

These are **very important for Python logic questions** because they teach % and //.

Last Digit

Last digit = Number % 10

```
last_digit = number % 10
```

Remove Last Digit

Number without last digit = Number // 10

```
number = number // 10
```

Example: Three-Digit Number

For num = 583:

Hundreds digit:

```
hundreds = num // 100
```

Tens digit:

```
tens = (num // 10) % 10
```

Units digit:

```
units = num % 10
```

Sum of Three Digits

Sum = Hundreds + Tens + Units

```
sum_digits = hundreds + tens + units
```

Reverse Three-Digit Number

```
reverse = units * 100 + tens * 10 + hundreds
```

---

17. Even and Odd

Even Number

A number is even if:

Number % 2 == 0

```
if number % 2 == 0:
    print("Even")
```

Odd Number

Number % 2 != 0

```
if number % 2 != 0:
    print("Odd")
```

---

18. Divisibility

Divisible by 2

number % 2 == 0

Divisible by 3

number % 3 == 0

Divisible by 5

number % 5 == 0

Divisible by 10

number % 10 == 0

Divisible by Both 3 and 5

number % 3 == 0 and number % 5 == 0

---

19. Age-Based Logic

Adult

age >= 18

Between Two Ages

For age between 18 and 60:

age >= 18 and age <= 60

Or:

18 <= age <= 60

---

20. Basic Logical Conditions

AND

Both conditions must be true:

condition1 and condition2

Example:

age >= 18 and age <= 60

OR

At least one condition must be true:

condition1 or condition2

Example:

marks >= 40 or sports_quota == True

NOT

Reverses the result:

not condition

Example:

not (age < 18)

⭐ **Most Important Formulas for Your Students**

If these are **beginner Python students**, I would make sure they are comfortable with these **15 formulas first**:

1. a + b → Addition
2. a - b → Subtraction
3. a * b → Multiplication
4. a / b → Division
5. a // b → Quotient
6. a % b → Remainder
7. a ** 2 → Square
8. (obtained / total) * 100 → Percentage
9. SP - CP → Profit
10. CP - SP → Loss
11. (profit / CP) * 100 → Profit %
12. (discount / 100) * MP → Discount
13. length * breadth → Rectangle Area
14. 2 * (length + breadth) → Rectangle Perimeter
15. number % 2 == 0 → Even Number

These formulas are enough to create a **large number of arithmetic + logical operator + if/else practice questions** for beginners.

Mathematics Formulas Required for the Questions

1. **Sum** = a + b
2. **Difference** = a − b
3. **Product** = a × b
4. **Average** = Sum of Values / Number of Values
5. **Percentage** = (Obtained Value / Total Value) × 100
6. **Simple Interest** = (P × R × T) / 100
7. **Total Amount** = Principal + Simple Interest
8. **Area of Rectangle** = Length × Breadth
9. **Perimeter of Rectangle** = 2 × (Length + Breadth)
10. **Area of Circle** = $\pi \times \text{Radius}^2$
11. **Perimeter of Square** = 4 × Side
12. **Square of Number** = Number^2
13. **Cube of Number** = Number^3
14. **Discount Amount** = (Price × Discount%) / 100
15. **Final Price after Discount** = Price − Discount
16. **GST Amount** = (Price × GST%) / 100
17. **Final Price after GST** = Price + GST
18. **Profit** = Selling Price − Cost Price
19. **Loss** = Cost Price − Selling Price

20. **Profit Percentage** = (Profit / Cost Price) × 100
21. **Loss Percentage** = (Loss / Cost Price) × 100
22. **Total Bill** = Price × Quantity
23. **Monthly Salary** = Annual Salary / 12
24. **Total Salary** = Basic Salary + Allowance
25. **Age** = Current Year − Birth Year
26. **Days to Hours** = Days × 24
27. **Hours to Minutes** = Hours × 60
28. **Minutes to Seconds** = Minutes × 60
29. **Kilometers to Meters** = Kilometers × 1000
30. **Meters to Centimeters** = Meters × 100
31. **Rupees to Paise** = Rupees × 100
32. **Weeks** = Total Days ÷ 7
33. **Remaining Days** = Total Days mod 7
34. **Celsius to Fahrenheit** = (Celsius × 9/5) + 32
35. **Fahrenheit to Celsius** = (Fahrenheit − 32) × 5/9
36. **Double** = Number × 2
37. **Triple** = Number × 3
38. **Remainder** = Dividend mod Divisor
39. **Quotient** = Dividend ÷ Divisor
40. **Electricity Bill** = Units Consumed × Price per Unit